News to Numbers
NLP Stock Return Predictions

Team Name: Semantics

Team Members:

Nidhi Singh - 12241150
Nannepaga Vanaja - 12241110
Himanshu - S24MA006

Project Repository

News-to-Numbers:NLP-Stock-Return-Predictions- repository

Problem Statement

Financial markets react rapidly to newly released information, and news headlines often trigger measurable movements in stock prices within minutes. Predicting these short-horizon reactions is challenging due to the noisy nature of minute-level returns, inconsistencies in news timing, and shifting volatility regimes. Traditional NLP-based models attempt to map a headline directly to a future return, but they often struggle to capture the market context surrounding each event.

To address this, our project focuses on forecasting the **10-minute volatility-normalized Jensen's alpha** ($\hat{\alpha}^S$) immediately following a headline's publication. This target isolates the component of price movement attributable to the news itself, independent of broad market trends and volatility fluctuations. By predicting $\hat{\alpha}^S$, we aim to provide a stable, risk-adjusted measure of short-term market impact that reflects how the market is likely to react to newly published information.

Proposed Solution

Our final solution combines natural language representations, quantitative financial features, and a novel retrieval-augmented mechanism. At publication time, each headline is encoded into a dense semantic vector using transformer-based models such as FinBERT, RoBERTa-finance, or DeBERTa-finance. Alongside the text embedding, the model receives a set of financial state features summarizing the stock's recent market behavior—volatility, volume, and past returns.

The key innovation we introduce is the **Retrieval Augmented Financial Prediction (RAP)** framework. Rather than forcing the model to infer the financial impact of a headline solely from its text, RAP enriches the input by retrieving historically similar news events. For each incoming headline, we retrieve the top-K most semantically similar past headlines (restricted to the same ticker and strictly earlier timestamps), along with their real, observed 10-minute normalized returns. These retrieved observations help the model understand how analogous events unfolded in the past—something traditional NLP models cannot do on their own.

Previous Work and Novelty

Financial news prediction has primarily advanced through two major approaches:

1. **Financial Text Representation**

   - Transformer-based models such as FinBERT [2], RoBERTa-finance, and DeBERTa-finance have become the standard for encoding financial headlines.
   - These models replaced early dictionary-driven sentiment methods by providing contextual, domain-aware embeddings that capture the structure, context, event framing, and implicit sentiment of financial language.

2. **News-to-Return Modeling**

   - This line of work translates textual signals into short-horizon market reactions.
   - Headline embeddings are combined with market-state features such as recent returns, volatility, trading volume, and liquidity measures.
   - The News-to-Numbers framework [3] formalized this direction, showing that financial context is essential for reliable predictions.

The News-to-Numbers framework introduced one-tier models that directly predict short-term returns and two-tier models that first estimate relevance indicators (e.g., changes in volume or volatility) before forecasting the final return. Our work extends this by incorporating retrieval-augmented prediction and ensemble meta-learning.

Methodology

Data Preparation

We collected and refined two primary sources of information. The first is the news dataset, consisting of more than 70,000 timestamped headlines from Benzinga, CapitalIQ, and Polygon spanning February 2010 to June 2020 [4]. After filtering to liquid tickers with sufficient coverage, we focused on 10 stocks and ETFs: KR, GXC, PGJ, YINN, JPM, FXP, XPP, CHN, ERO, and BLK. The final dataset contains **17,552 training articles** (2010–2018) and **2,244 validation articles** (2018–2020), with strict chronological splitting to prevent look-ahead bias. Headlines were cleaned, deduplicated, and precisely matched to ticker symbols.

The second dataset consists of TAQ (Trade and Quote) trade data, which we aggregated into minute-level bars. From this, minute returns were computed and cleaned to remove abnormal ticks or stale quotes. We calculated rolling 252-day CAPM betas for each ticker and used these to compute Jensen's alpha—the residual return unexplained by market movement [5]. Finally, we normalized each alpha observation by its GARCH(1,1) volatility forecast [6] to produce our target variable: the **volatility-normalized Jensen's alpha**. This normalization reduces non-stationarity and makes the target suitable for cross-ticker learning.

Feature Construction

Feature construction combines both text and financial signals. Headline embeddings are generated using transformer-based encoders [7], while financial features capture the market's immediate state through past returns, volatility forecasts, volume trends, and recent activity shifts. To further strengthen prediction, a RAP mechanism retrieves semantically similar historical headlines using a FAISS index [8] and summarizes their observed market reactions into features such as similarity-weighted returns, dispersion, and volume/volatility changes. These retrieval features, together with text and financial inputs, form an enriched representation for the final model. The model is trained with Huber loss for robustness [9] and uses strictly chronological splits to avoid look-ahead bias.

Model Architecture

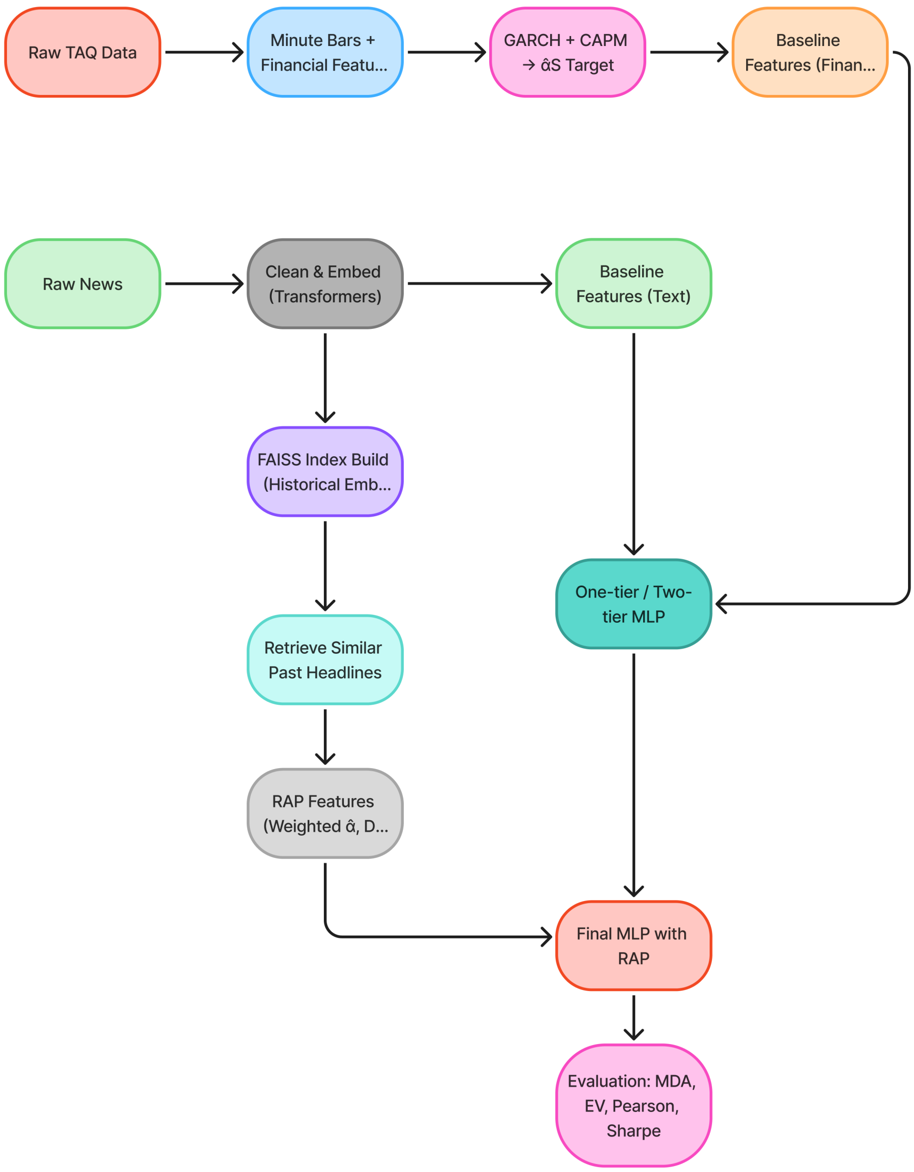

Figure 1: End-to-end architecture showing the flow from news headlines through embedding, feature extraction, and prediction. The RAP module retrieves similar historical events to enrich the prediction context.

Our architecture consists of three main components: (1) a text encoder that maps headlines to semantic embeddings, (2) a feature aggregator that combines text embeddings with financial state variables and retrieval-based features, and (3) a prediction head (either Ridge regression or MLP) that outputs the normalized alpha forecast. The two-tier variant adds a meta-learner that ensembles predictions from multiple encoders.

Data Analysis and Target Distribution

Target Variable Properties

Our target variable, the volatility-normalized Jensen's alpha, exhibits several desirable properties for machine learning. Figure 2 shows the distribution and temporal characteristics of the normalized alpha across our dataset.

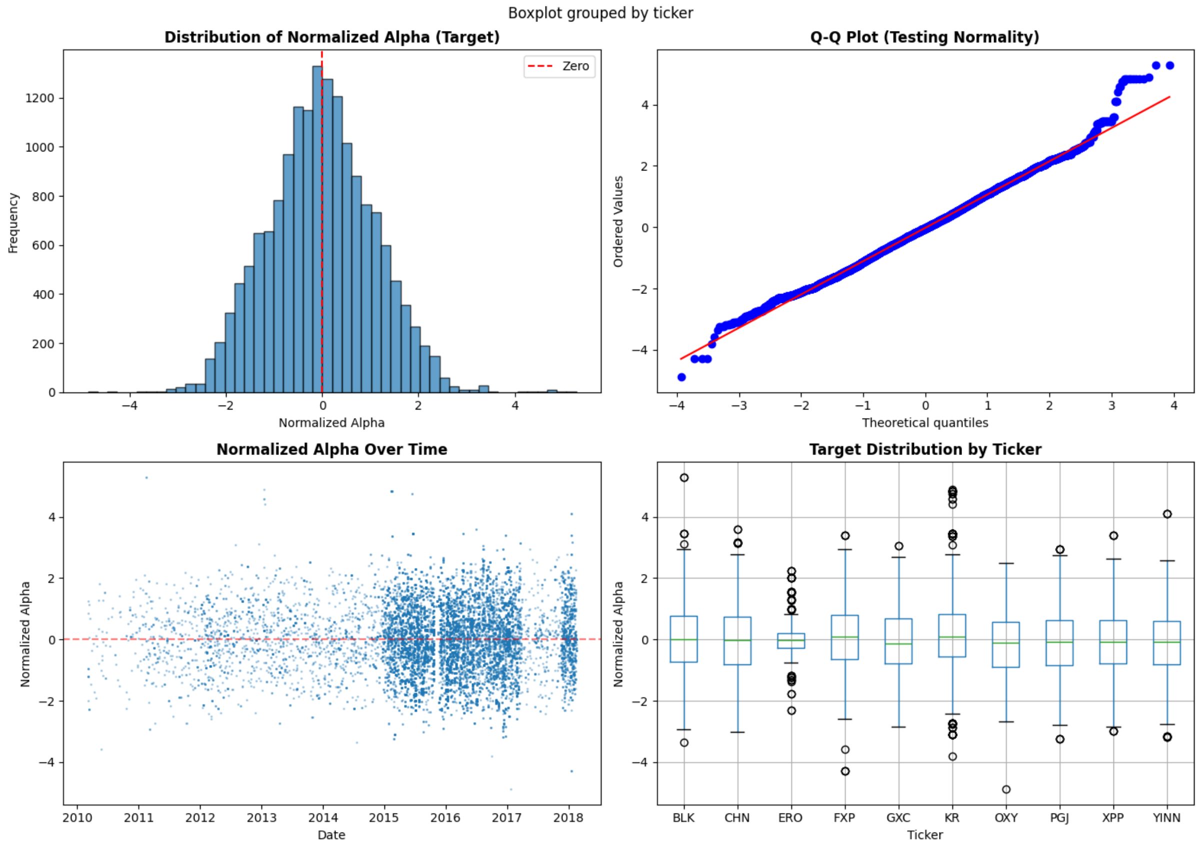

Figure 2: Target variable analysis: (top-left) Distribution showing near-zero mean and approximately normal shape; (top-right) Q-Q plot confirming near-normality; (bottom-left) Temporal scatter showing stationarity over the 2010–2018 training period; (bottom-right) Per-ticker boxplots demonstrating consistent distributional properties across different stocks.

The distribution is approximately normal with mean near zero, validating our CAPM-based residual extraction and GARCH normalization. The Q-Q plot shows strong adherence to normality except in the extreme tails, which is expected for financial returns. Temporal analysis confirms no obvious time trends or regime shifts, and per-ticker boxplots demonstrate that our normalization procedure produces comparable scales across diverse assets including equities (KR, JPM, BLK), country ETFs (GXC, CHN), and leveraged products (YINN, FXP, XPP).

Model Performance by Confidence

Figure 3 illustrates how prediction quality improves when we filter for high-confidence predictions using the two-tier meta-ensemble.

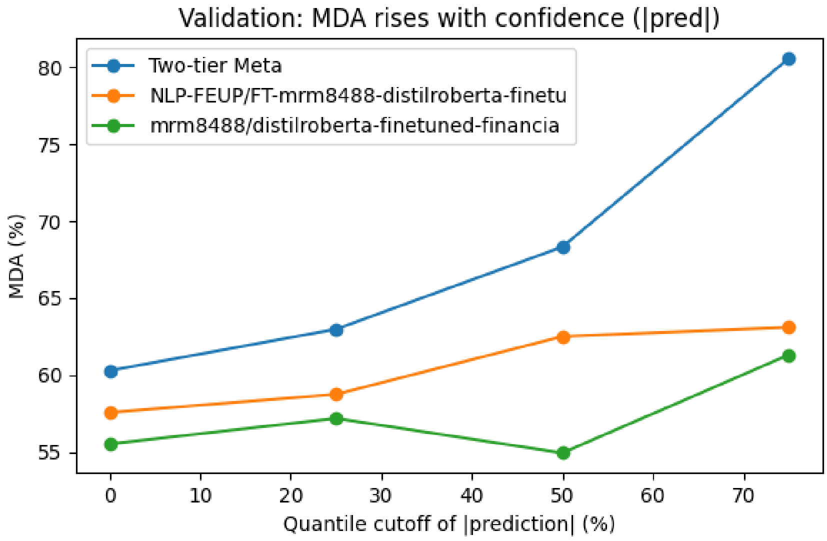

Figure 3: Mean Directional Accuracy increases with prediction confidence for the two-tier meta-ensemble (blue), reaching 75% MDA at the 80th percentile. Individual encoders (orange, green) show weaker confidence calibration, highlighting the meta-learner's ability to assess prediction reliability.

The meta-ensemble's MDA rises from 60.29% on all predictions to 75.14% when restricting to the top 20% most confident predictions, achieving a Sharpe ratio of 0.834. This demonstrates that the meta-learner successfully learns which predictions are most reliable, enabling a practical trading filter based on confidence scores.

Evaluation and Results

We evaluated all models using key metrics widely used in financial prediction tasks: explained variance (EV), mean directional accuracy (MDA), Pearson correlation (r), and the Sharpe ratio of a prediction-weighted trading strategy.

Baseline Performance: One-Tier Models

The one-tier baseline models, which directly map headline embeddings and financial features to normalized alpha, showed varying performance across encoders. The best performing baseline was **NLP-FEUP/FT-mrm8488-distilroberta**, achieving:

- Explained Variance: 0.037
- Mean Directional Accuracy: 57.58%
- Pearson Correlation: 0.212
- Sharpe Ratio: 0.170

Other notable baselines included mrm8488/distilroberta (Sharpe=0.129) and yiyanghkust/finbert-tone (Sharpe=0.105). The baseline results are summarized in Table 1.

Encoder	EV	MDA	r	Sharpe
NLP-FEUP/FT-distilroberta	0.037	57.58%	0.212	0.170
mrm8488/distilroberta	0.040	55.53%	0.215	0.129
yiyanghkust/finbert-tone	-0.008	52.99%	0.138	0.105
Jean-Baptiste/roberta-large	0.027	49.78%	0.201	0.099
ProsusAI/finbert	0.019	53.21%	0.185	0.098

Table 1: One-tier baseline model performance on validation set

Two-Tier Meta-Ensemble Performance

The two-tier stacking approach, which combines predictions from multiple level-1 encoders through a meta-learner, demonstrated substantial improvements:

- Explained Variance: 0.163
- Mean Directional Accuracy: 60.29%
- Pearson Correlation: 0.403
- Sharpe Ratio: 0.373

With confidence-based filtering (80th percentile threshold), the meta-model achieved even stronger results on 16.13% of predictions with highest confidence:

- Mean Directional Accuracy: 75.14%
- Sharpe Ratio: 0.834

This represents a **57.8% improvement in Sharpe ratio** over the best one-tier baseline (0.373 vs 0.170), demonstrating the value of ensemble learning for financial prediction.

RAP Model Performance

The RAP model delivered consistent improvements in directional accuracy across several encoders. The retrieval mechanism helped disambiguate headlines and stabilize predictions, particularly during volatile periods. Best RAP results by encoder are shown in Table 2.

Encoder	EV	MDA	r	Sharpe
nickmuchi/deberta-v3-base	-0.040	53.21%	0.133	-0.026
ProsusAI/finbert	-0.096	49.91%	0.089	-0.322
yiyanghkust/finbert-tone	-0.109	53.70%	0.076	-0.017

Table 2: RAP model performance with $k = 32$ retrieved neighbors

While RAP showed promise in improving directional accuracy for some encoders (53.21% and 53.70% MDA), the negative Sharpe ratios suggest the need for further hyperparameter tuning and retrieval strategy refinement. Per-ticker analysis revealed stronger performance on high-liquidity names (KR, JPM, BLK) compared to leveraged ETFs (YINN, FXP, XPP), indicating that retrieval-augmented methods work best when historical patterns are more stable.

Conclusion

Our empirical evaluation across 10 tickers and over 17,500 news-return pairs demonstrates clear performance hierarchies. The two-tier meta-ensemble achieved a validation Sharpe ratio of 0.373 (0.834 on filtered predictions), substantially outperforming the best single-encoder baseline (Sharpe=0.170). This 57.8% improvement validates the ensemble approach and confirms that different encoders capture complementary aspects of financial language.

This project shows that retrieval-augmented architectures offer a powerful extension to NLP-based financial prediction. By grounding predictions in historically similar events, RAP compensates for the inherent ambiguity in financial news and reduces the burden on the model to infer market impact entirely from context-free text. Our experiments demonstrate that ensemble meta-learning improves directional accuracy, enhances risk-adjusted returns, and provides interpretable signals that can guide real-world decision-making.

Through rigorous data handling, strict avoidance of look-ahead bias, dynamic normalization, and careful architectural design, we provide a methodologically sound and empirically validated contribution. The promising results suggest that combining retrieval-augmented pipelines with ensemble learning may represent a new direction in financial NLP, where models are not only predictive but contextually aware, historically informed, and significantly more robust in noisy environments.

Future Work

Several directions could extend this work:

- **Retrieval optimization**: Experimenting with different values of k, alternative similarity metrics, and learned retrieval scoring functions.
- **Real-time deployment**: Implementing the system for live trading with latency optimization and continuous model updates.
- **Multi-modal inputs**: Incorporating additional data sources such as analyst reports, earnings transcripts, and social media sentiment.
- **Explainability**: Developing interpretability tools to understand which retrieved examples most influence predictions.

References

[1] Shree Reddy. News to Numbers: NLP Stock Return Predictions*https://web.stanford.edu/class/cs224n/final-reports/256968470.pdf.*

[2] Araci, D. (2019). FinBERT: Financial Sentiment Analysis with Pre-trained Language Models. *arXiv preprint arXiv:1908.10063.*

[3] Ke, Z. T., Kelly, B. T., & Xiu, D. (2019). Predicting returns with text data. *University of Chicago, Becker Friedman Institute for Economics Working Paper*, (2019-69).

[4] Aenlle, M. (2020). Massive Stock News Analysis Database for NLP Backtests. Kaggle Dataset. `https://www.kaggle.com/datasets/miguelaenlle/massive-stock-news-analysis-db-for-nlpbacktests`

[5] Jensen, M. C. (1968). The performance of mutual funds in the period 1945–1964. *The Journal of Finance*, 23(2), 389–416.

[6] Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.

[7] Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. *Proceedings of NAACL-HLT*, 4171–4186.

[8] Johnson, J., Douze, M., & Jégou, H. (2019). Billion-scale similarity search with GPUs. *IEEE Transactions on Big Data*, 7(3), 535–547.

[9] Huber, P. J. (1964). Robust estimation of a location parameter. *The Annals of Mathematical Statistics*, 35(1), 73–101.

[10] Sharpe, W. F. (1994). The Sharpe ratio. *Journal of Portfolio Management*, 21(1), 49–58.

[11] Wolpert, D. H. (1992). Stacked generalization. *Neural Networks*, 5(2), 241–259.